Niceness packages: the symmetric cube and quartic, finished

Samuel Lavery*

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Abstract

The finish of benchmark rungs 4–5, in the shape the architecture prescribes. A *niceness package* for a coefficient bank is an entire continuation carrying the prescribed $\Gamma_{\mathbb{C}}$-chart on a half-plane and the self-dual functional equation — exactly what classical analytic theory supplies per rank, consumed as typed data the way the Hecke eigenform package is. We prove: the continuation in a package is *unique*; packages *compose* (product bank, concatenated chart, product root number); the *rank-1 package is inhabited* — Hecke's completed functional equation with its entire transform, from the compiled seed rung, so the class is non-vacuous by construction; and, given the classical Sym^3 and Sym^4 packages of Kim–Shahidi (with Shimura's for Sym^2), the tensor objects $\mathrm{Sym}^3 \oplus \mathrm{std}$ and $\mathrm{Sym}^4 \oplus \mathrm{Sym}^2$ are fully nice, with the ζ-dressing of the degree-9 tensor the compiled Λ_ζ and its booked poles. Together with the uniform-transport theorem, every representation datum whose constituents carry packages is nice — the classical inputs isolated to named per-rank slots, ranks 1 and 2 discharged.

1 Packages

For a coefficient bank a, shifts (μ_j), threshold B, and root number ε, a *niceness package* consists of $\Lambda\colon \mathbb{C} \to \mathbb{C}$ entire with $\Lambda(1-s) = \varepsilon\,\Lambda(s)$ for all s and $\Lambda(s) = \prod_j \Gamma_{\mathbb{C}}(s+\mu_j)\cdot L(a,s)$ on $\Re s > B$ (`NicePackage`).

Theorem 1 (uniqueness)**.** *Two packages over the same chart data have the same transform: the chart pins them on an open half-plane and entirety globalizes by the identity theorem. (`NicePackage.unique`.)*

Theorem 2 (composition)**.** *Packages multiply: the product is a package for the convolution bank at the concatenated chart, larger threshold, and product root number (`NicePackage.mul`).*

Theorem 3 (the rank-1 anchor, inhabited)**.** *The seed standard rung packages: Λ the compiled entire transform of the weight-$2n$ eigenform, $\varepsilon = (-1)^n$, chart $\Gamma_{\mathbb{C}}(s + (n - \frac{1}{2}))\, L(c_{\mathrm{Sym}^1}, s)$ — Hecke's completed functional equation, from modularity alone. (`seedNicePackage`.)*

2 The finish

Theorem 4 (rung 4)**.** *Given a Sym^3 package (Kim–Shahidi, cited at published strength, consumed as typed data), the degree-6 tensor object $\mathrm{Sym}^3 \oplus \mathrm{std}$ is fully nice: entire, charted at the*

*Machine-checked companion: `RequestProject/NiceClosure.lean`, 6 named declarations, each at axiom footprint {`propext`, `Classical.choice`, `Quot.sound`} or a subset, no `sorry`, no `axiom`, zero `sorryAx`.

concatenated shifts, self-dual with root number $\varepsilon_3 \cdot (-1)^n$ — the standard factor discharged by the anchor. Its functional equation is exactly the collapsed wall's tensor side. (`tensor23_nice`, `tensor23_FE_of_package`.)

Theorem 5 (rung 5)**.** *Given* Sym^4 *and* Sym^2 *packages (Kim–Shahidi; Shimura), the paired product* $\mathrm{Sym}^4 \oplus \mathrm{Sym}^2$ *is fully nice; the residual ζ-dressing of the degree-9 tensor is the compiled Λ_ζ with its booked poles. (`sym4_sym2_nice`.)*

Addendum — the cube from one identity. The Sym^3 slot is upgraded past the package: in `RequestProject/Sym3ThetaWall.lean` the classical input is a *single pointwise identity* between two explicit theta functions — the self-duality $\phi_3(1/x) = \varepsilon\, x\, \phi_3(x)$ of the prescribed Sym^3 readouts at the pure-$\Gamma_{\mathbb{C}}$ chart $(n - \frac{1}{2}, 3(n - \frac{1}{2}))$ — and the entire analytic superstructure is machine-derived: the strong pair assembles with every other field compiled (`sym3Source`), it is its own contragredient by the clock-reversal bank identity and the identity theorem (`symrPair_dual_eq_primal`, `sym3_symmLambda_eq`), and continuation, entirety, functional equation, and chart all follow (`sym3_package_of_theta`), whence the full degree-6 tensor conclusion from the one identity (`tensor23_nice_of_theta`). The even rank Sym^4 carries a $\Gamma_{\mathbb{R}}$-factor outside the pure-$\Gamma_{\mathbb{C}}$ kernel. Its one-identity finish at the mixed chart is now compiled: `sym4_package_of_theta` in `ThetaMechanism.lean` supersedes the package slot.

Remark 6 (register and falsifiers)**.** *Proven, machine-checked, unconditional: the package calculus (uniqueness, composition), the inhabited rank-1 anchor, and both tensor conclusions given their per-rank packages. Cited, consumed as typed data at named slots: Kim–Shahidi at $r = 3, 4$ and Shimura's holomorphy at $r = 2$ — the same status the Hecke package has throughout this series. Compiled elsewhere in the series: the* Sym^2 *reflection (the lattice route), the Euler determinations pinning* $L(\mathrm{Sym}^3)$ *and* $L(\mathrm{Sym}^4)$*, and the wall collapses making each typed slot exactly one self-dual identity. A counterexample would be two packages over one chart with different transforms, or a product violating the composed chart — each formally stated, so a counterexample would require a kernel failure.*

References

[1] H. Kim and F. Shahidi, *Functorial products for* $\mathrm{GL}(2) \times \mathrm{GL}(3)$ *and the symmetric cube for* $\mathrm{GL}(2)$, Ann. of Math. **155** (2002), 837–893; H. Kim, *Functoriality for the exterior square of* GL_4 *and the symmetric fourth of* GL_2, J. Amer. Math. Soc. **16** (2003), 139–183.

[2] G. Shimura, *On the holomorphy of certain Dirichlet series*, Proc. London Math. Soc. **31** (1975), 79–98.